DevFlow AI

Intelligent Task Management System

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The Problem We're Solving

The Challenge

Traditional project management systems depend on manual planning and subjective time estimations. This often results in inaccurate scheduling, missed deadlines, and reduced team productivity.

Our Solution

DevFlow AI leverages Artificial Intelligence and Machine Learning to automate workflow planning. The system provides a smart dashboard that accurately predicts task duration and automatically generates detailed technical steps.

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Three-Layer Architecture

DevFlow AI employs a robust three-tier architecture that separates concerns and ensures scalability.

Presentation Layer

Django Templates with responsive UI components that adapt to any device screen size.

Intelligence Layer

Scikit-learn for machine learning predictions and Gemini 2.0 for generative AI capabilities.

Data Layer

SQLite database with Django ORM for structured, persistent data storage and retrieval.

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Machine Learning Integration

Linear Regression Model

Our ML model uses Scikit-learn to predict task duration based on complexity levels ranging from 1 to 5. The algorithm applies the linear equation $y = mx + c$ to map complexity scores to estimated hours.

This supervised learning approach improves accuracy over time as more task data becomes available for training.

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Complexity Levels

Simple to highly complex tasks

24

Hour Range

From 1 to 24+ hours predicted

100%

Automation

Zero manual time entry needed

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Gemini 2.0: Generative AI Power

Gemini 2.0 Flash API

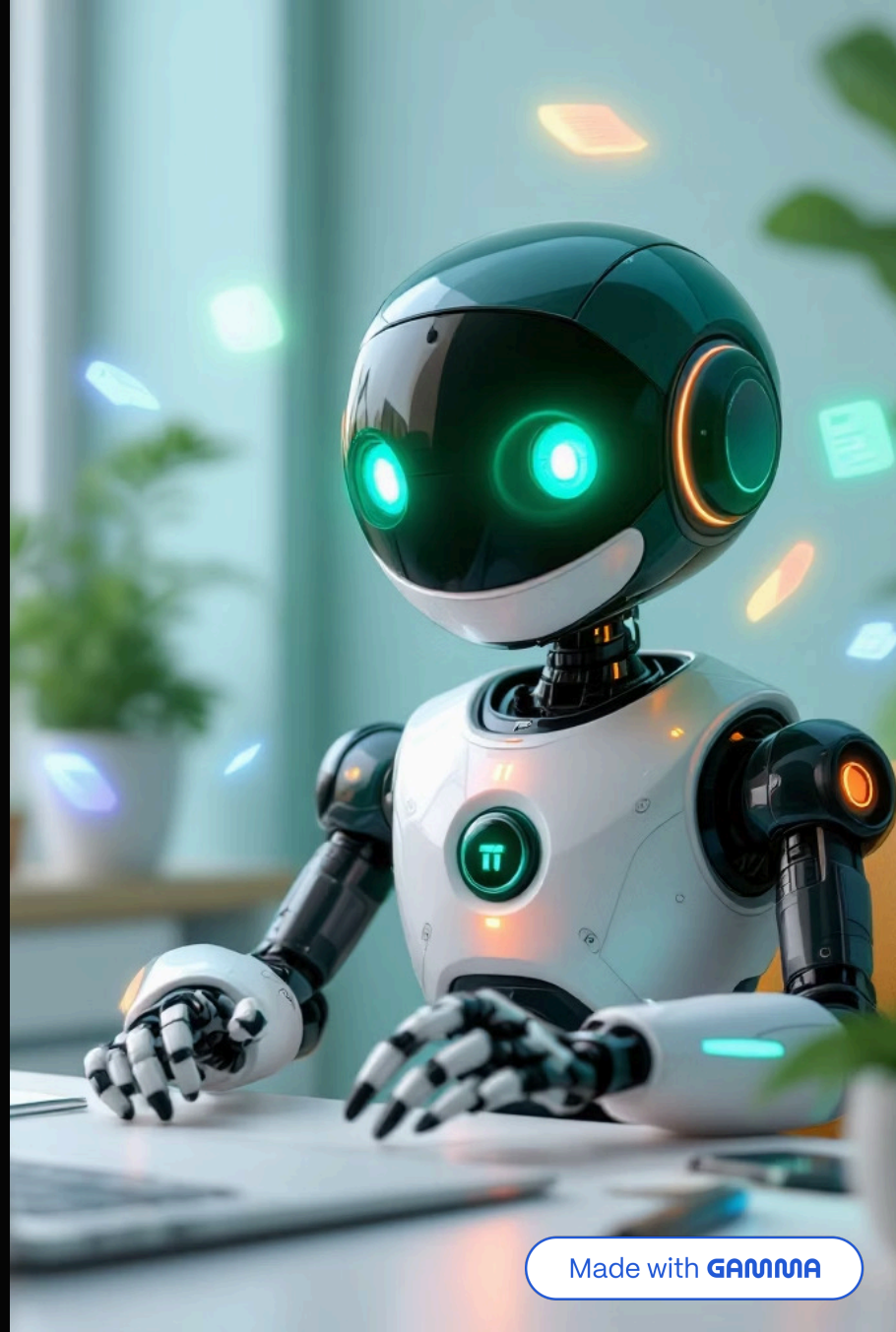
Google's latest generative AI model for fast, accurate technical breakdowns.

Auto-Task Generation

Breaks complex features into 3 actionable technical sub-tasks instantly.

Time Savings

Reduces manual effort in technical documentation and planning phases.



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Database Design & Security

SQLite3 with Django ORM

The database stores user credentials, task details including complexity scores, estimated hours, and AI-generated sub-tasks. Django's Object-Relational Mapping simplifies database operations.

Key Tables

- Users (credentials and roles)
- Tasks (complexity, hours, AI sub-tasks)
- Projects (task groupings and status)

Role-Based Access Control

RBAC ensures only administrators access the management panel, while regular users view their assigned tasks. This security layer protects sensitive project data.

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Development Timeline

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Phase 1: Foundation

Project setup, database schema design, and Django configuration with URL routing.

2

Phase 2: Intelligence

ML model training with sample data, Gemini API integration for task breakdown generation.

3

Phase 3: Interface

Frontend UI development with responsive templates and comprehensive unit testing.

4

Phase 4: Deployment

Final system testing, documentation completion, and production deployment.

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Key Features & Benefits

Accurate Time Predictions

ML-powered duration estimates based on historical complexity patterns reduce planning errors.

Automated Task Breakdown

Gemini AI automatically generates 3 technical sub-tasks from complex features.

Smart Dashboard

Centralized view of all tasks, predictions, and progress tracking in real-time.

Secure Access

Role-based permissions ensure data protection and appropriate access levels.

Self-Improving Model

Linear regression model accuracy increases with more training data over time.

Responsive Design

Works seamlessly across desktop, tablet, and mobile devices with adaptive layouts.

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Technology Stack

Backend Framework

- Django (Python web framework)
- SQLite3 database
- Django ORM for data modeling

AI/ML Components

- Scikit-learn (Linear Regression)
- Google Gemini 2.0 Flash API

Frontend Technologies

- Django Templates (HTML/CSS)
- Responsive UI design

Security Features

- Role-Based Access Control
- Data validation and sanitization

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Future Roadmap

Advanced ML Models

Upgrade to ensemble methods and neural networks for higher prediction accuracy.

Real-Time Collaboration

Add WebSocket support for team coordination and live updates.

Cloud Deployment

Containerize with Docker, deploy on cloud platforms using Nginx production server.

Project Summary

DevFlow AI demonstrates how modern AI and ML technologies can significantly enhance developer productivity and project management. By automating time estimation and task breakdown, the system reduces manual planning effort while improving accuracy and team efficiency.

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